



Interest Rate Risk Flow View

Case 1 C – Group 3
Financial Engineering

The real question for treasuries isn't how much cash or cash flow they have today.

It is how exposed that cashflow is to tomorrow's uncertainty.

Presentation Roadmap

Our Goal is to help you understand, *"How much cash flow risk can we accept, how to identify it, and what's the price to hedge it?"*

- 1 The Challenge
- 2 Sources of Risk
- 3 What is Cash Flow at Risk?
- 4 Interest Rate Scenarios
- 5 Hedging Trade-offs
- 6 Calibration Choices
- 7 Recommendations

The Challenge

With rates more volatile than at any time in the past decade, treasurers face a multiple challenges including refinancing at uncertain costs, absorbing higher floating debt expenses, and navigating unpredictable reinvestment opportunities.

These challenges can be categorized into

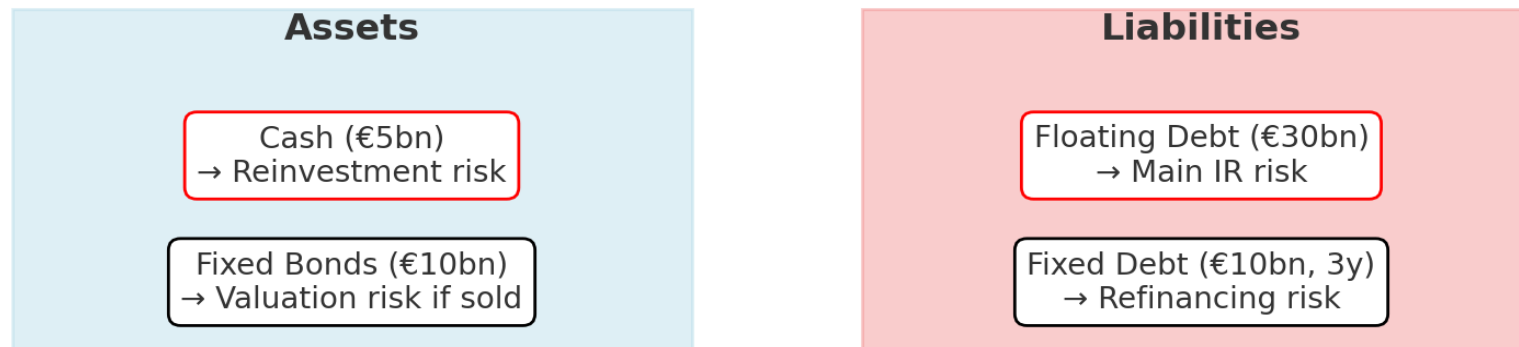
- **Uncertainty:** Rate environment is more volatile than the past decade.
- **Complexity:** Multiple exposures across different time horizons.
- **Planning:** Budget cycles need probabilistic risk assessment.
- **Stakeholders:** CFO & board demand clear, forward-looking communication.

Where does cash flow volatility come from?

- **Revenue** → FX exposure, demand risk, commodity-linked sales,
- **Operating Expenses** → energy/raw material price volatility
- **Assets** → cash (reinvestment risk), fixed bonds (market value risk), customer payment terms
- **Liabilities** → floating-rate debt (Interest rate exposure), fixed-rate debt (refinancing risk), supplier payment terms

A Reference Example

Sources of Cash Flow Risk - Balance Sheet View



Concept of Cashflow at Risk

CFaR or Cash Flow at Risk measures the potential adverse impact on future cash flows due to interest rate movements over a specific time horizon at a given confidence level.

$$CFaR_{\alpha} = \text{Expected Cash Flow} - \text{Worst Case Cash Flow (at } \alpha \text{ level)} = E(CF) - Q_{\{\alpha\}}(CF)$$

▶ Where : α = confidence level (e.g., 95%); Q = quantile of cash flow distribution

Lets understand this with a simple question,

“What if interest rates suddenly rise by 1% — how much more will our company pay in interest?”

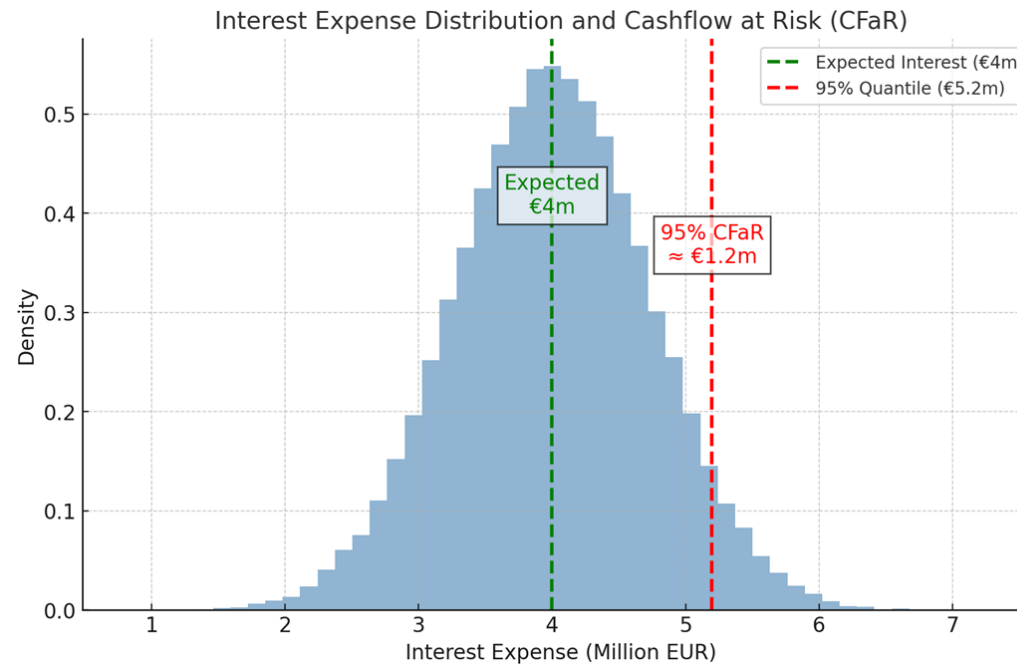
$$CFaR_{\{95\%\}} = E(CF) - Q_{\{95\%\}}(CF \mid \Delta r = 1\%)$$

▶ Where : r = Interest Rate; $\Delta r = 1\%$ (Interest rate increase)

One Simple Example

Let's assume a floating loan of notional € 100 M

- Expected interest at 4% = €4m
- Let's say that after simulating multiple interest rate change scenarios, we find that With 95% probability, annual interest expense will not exceed €5.2m. In 5% of cases, it could be higher



“With 95% confidence, interest expense will not exceed €5.2m, which is still €1.2m worse than the budget”

This is not just an abstract number. It tells the treasurer: keep at least €1.2m of additional liquidity in reserve or consider hedging

Advantages & Disadvantages

Approach	Advantages	Disadvantages	Difference vs. CFaR
Duration/Convexity	Simple, intuitive measure of interest rate sensitivity	Ignores distribution and tail risk	CFaR captures full distribution in cash and tail risk, not just sensitivity
Value at Risk (VaR)	Widely used, regulator-recognized, strong comparability	Focuses on portfolio value rather than liquidity	CFaR focuses on liquidity and budget certainty
Earning at Risk (EaR)	Linked to accounting earnings, aligns with financial statements	May underestimate liquidity stress	CFaR highlights cashflow pressure beyond reported earnings
Scenario/Stress test	Flexible, easy to communicate "what-if" stories	Depend on assumptions, no probability meaning	CFaR provides a probabilistic, confidence-level based risk measure

CFaR Pros

- Dollarized measure: Expressed directly in cash (extra potential outflow)
- Liquidity-centric
- Probabilistic Measure
- Captures market risk drivers (rates, FX, etc)

CFaR Cons

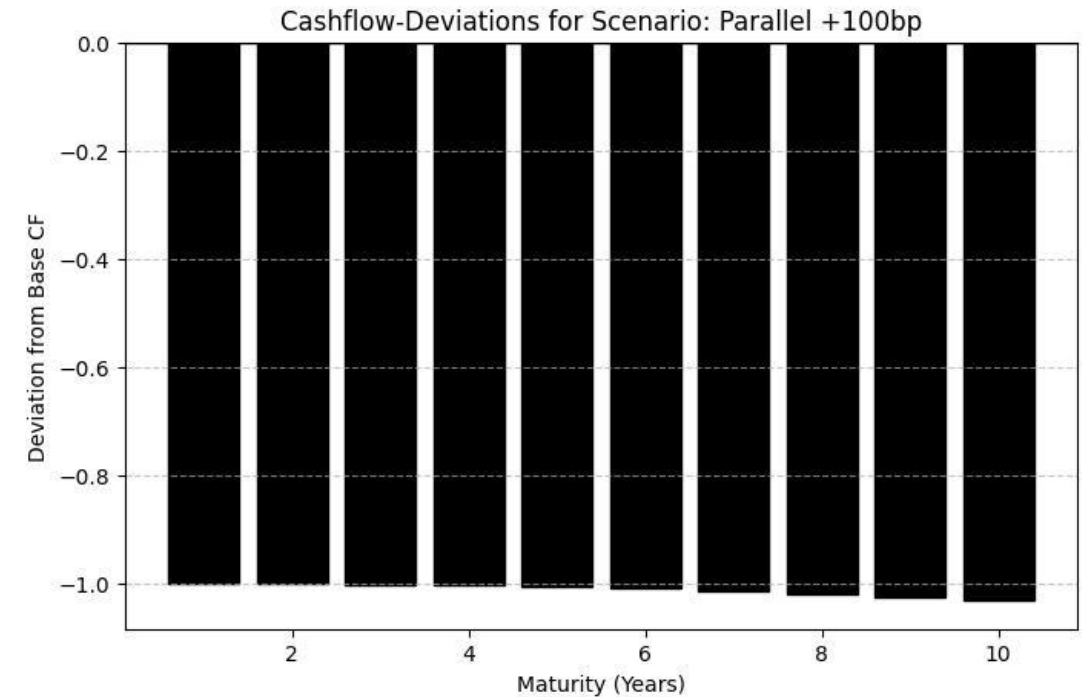
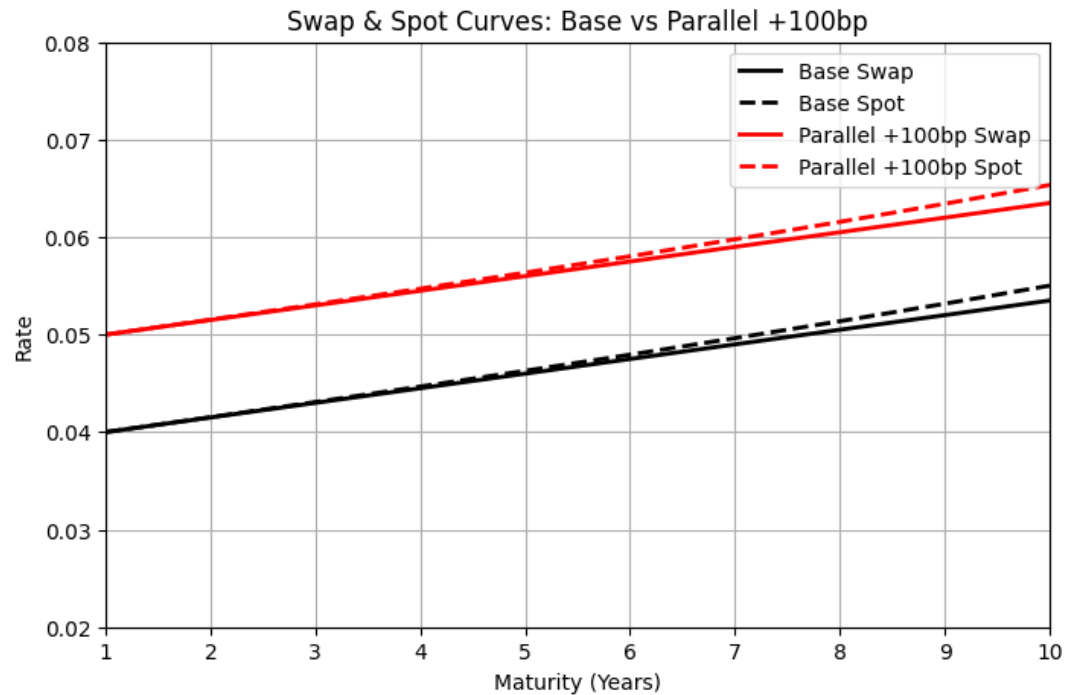
- Model and calibration dependent
- Data Heavy
- Not regulator-standardized

Why Cash Flow at Risk Matters

Multiple exposures, different timings, various risk types → We need integrated measurement

- Each risk source creates unique uncertainty in future cash inflows/outflows.
- Traditional tools like duration, sensitivities, and simple shocks provide useful but partial views — they don't capture probabilities.
- CFaR, like VaR, is a probabilistic measure but focused on future cash flows instead of portfolio values.
- CFaR integrates everything into one probabilistic metric: *“With 95% confidence, net cash flows will not be worse than €X.”*

Interest Rate Scenario 1: Parallel +100bp



Cash Flow Impact on the Asset Side

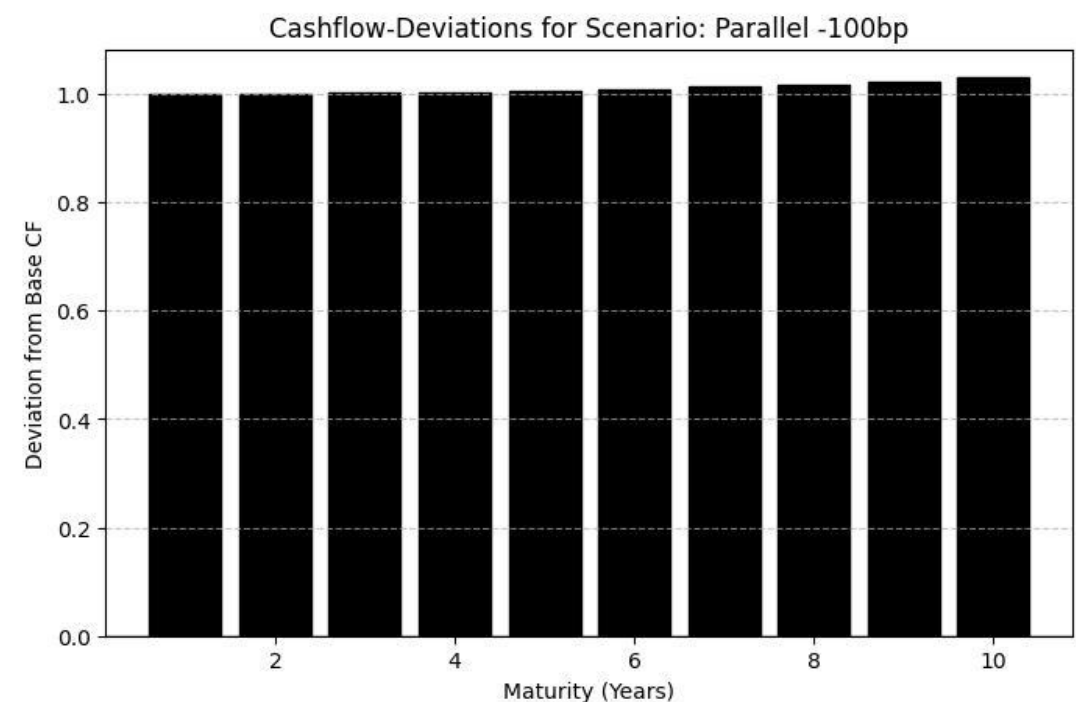
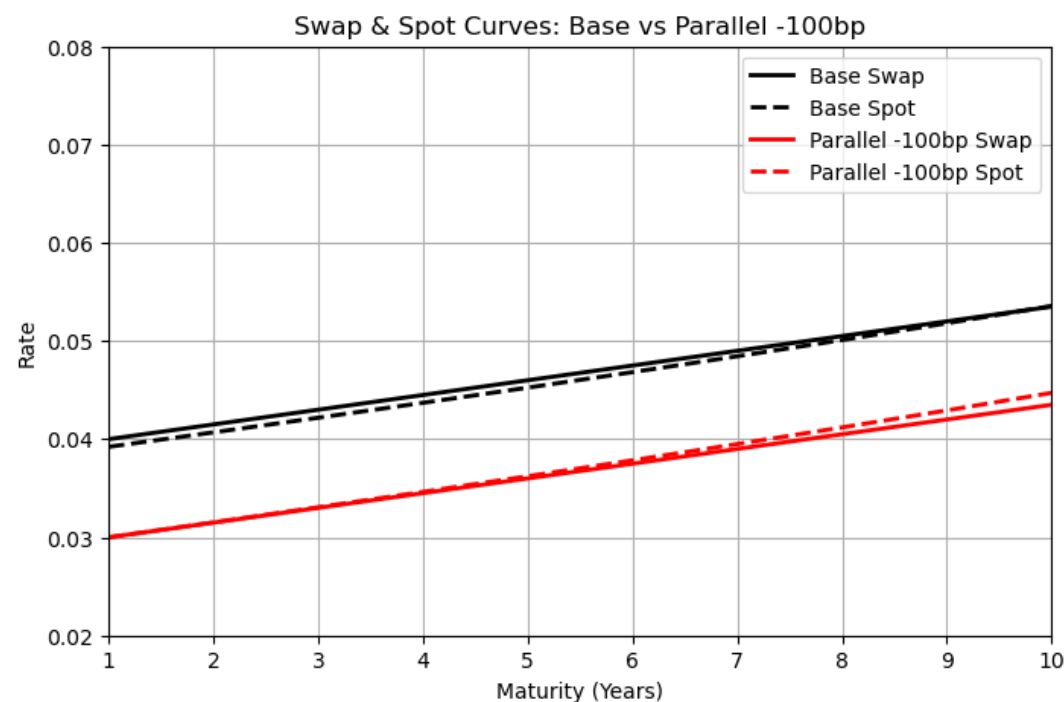
- Higher interest income from floating-rate investment
- Higher interest income for investment rollover
- Decreased value of fixed-rate investment, no cash flow impact

Cash Flow Impact on the Liability Side

- Higher interest expense for floating-rate liabilities
- Higher interest expense for refinancing
- No cashflow impact from value depreciation of fixed-rate liabilities

Direction of net cashflow effect depending on Asset/Liability sensitivity

Interest Rate Scenario 2: Parallel - 100bp



Cash Flow Impact on the Asset Side

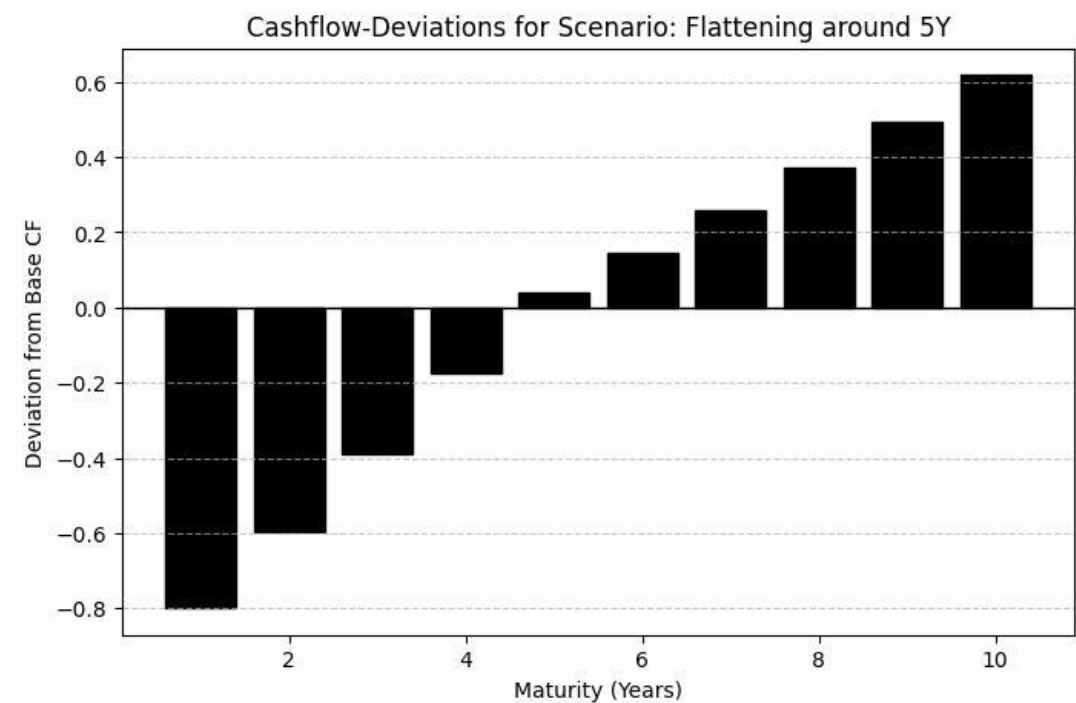
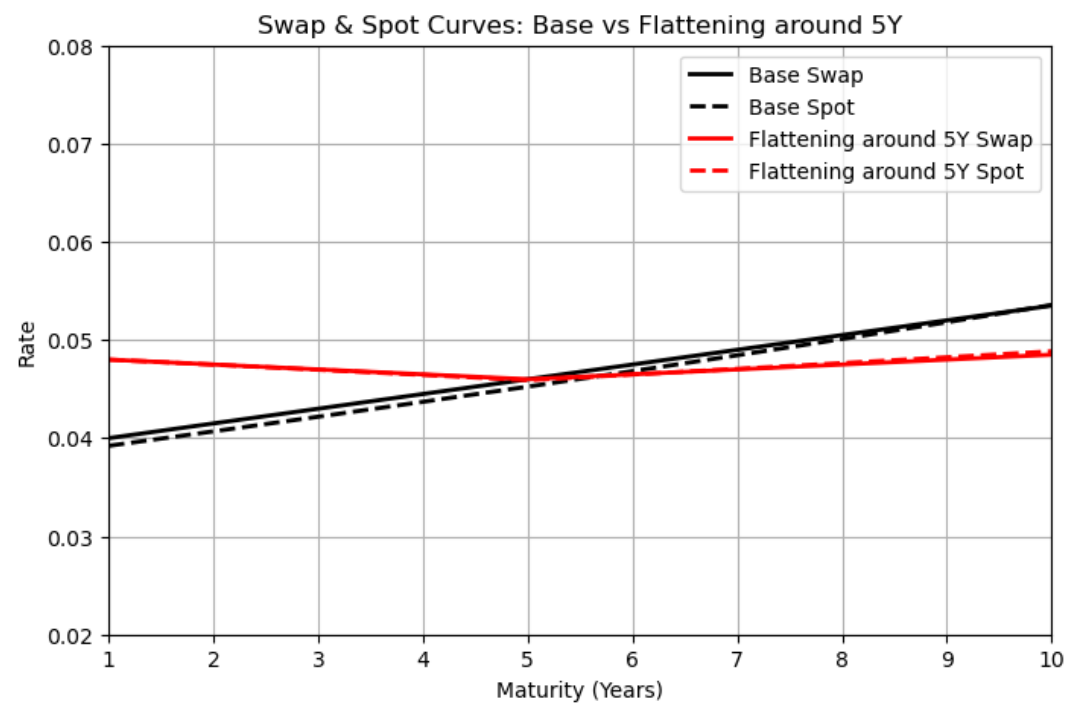
- Lower interest income from floating-rate investment
- Lower interest income for investment rollover
- Increased value of Long-Term investment, no cash flow impact

Cash Flow Impact on the Liability Side

- lower interest expense for floating-rate liabilities
- Lower interest expense for refinancing
- No cashflow impact from value appreciation of fixed-rate liabilities

Direction of net cashflow effect depending on Asset/Liability sensitivity

Interest Rate Scenario 3: Flattening the Curve Around 5Y



Cash Flow Impact on the Asset Side

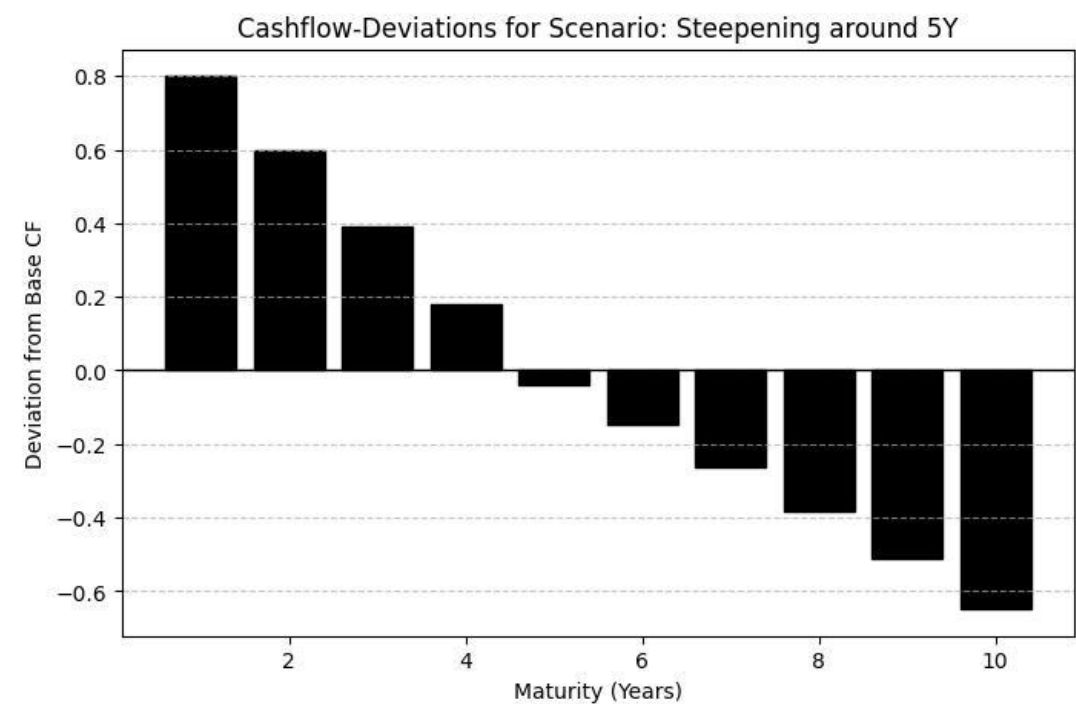
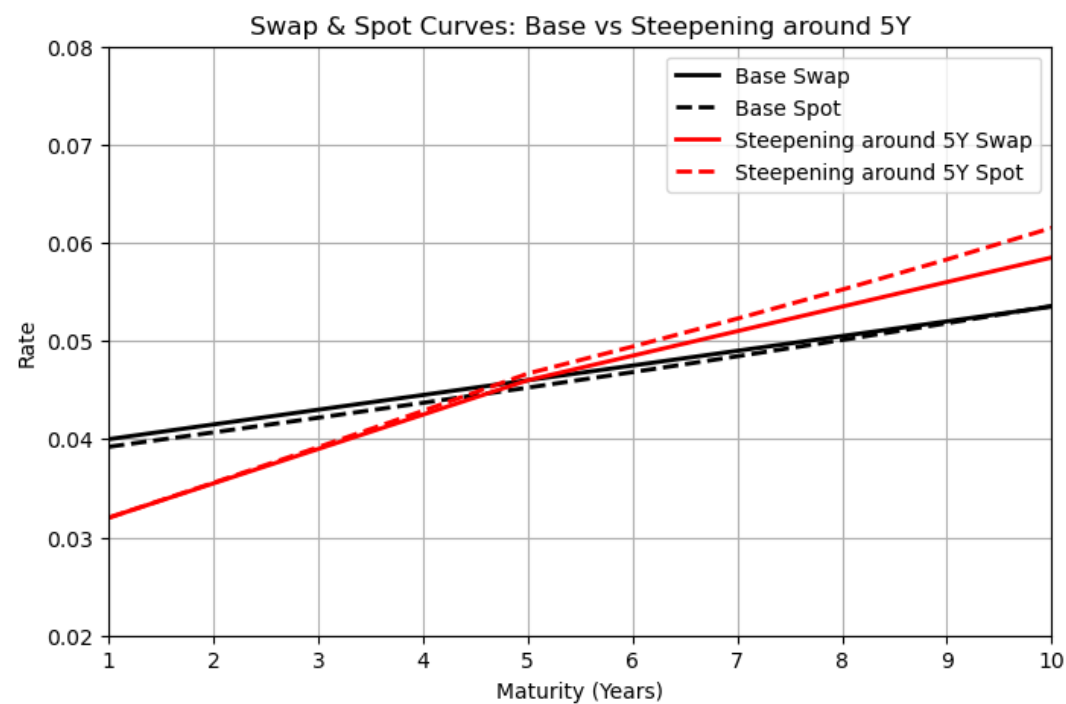
- Higher interest income from Short-Term investment
- Higher interest income for investment rollover
- Increased value of Long-Term investment, no cash flow impact

Cash Flow Impact on the Liability Side

- Higher interest expense for Short-Term liabilities
- Lower interest expense for Long-Term refinancing
- No cashflow impact from value appreciation of Long-Term liabilities

Positive net cashflow effect given Short-Term Asset and Long-Term Liabilities

Interest Rate Scenario 4: Steepening the Curve Around 5Y



Cash Flow Impact on the Asset Side

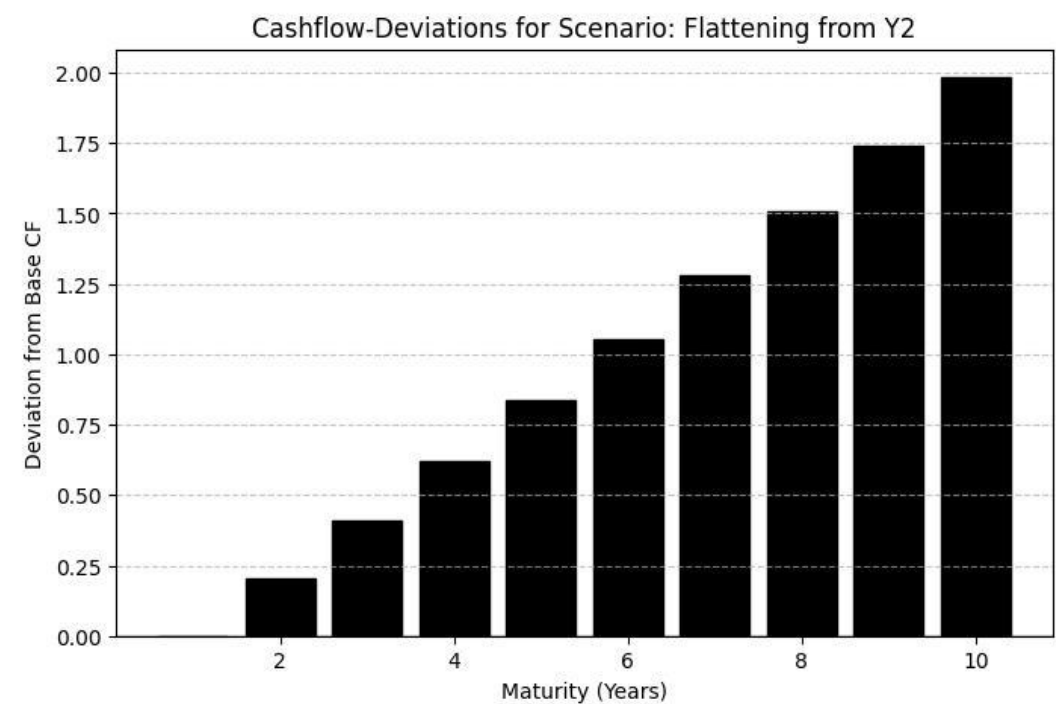
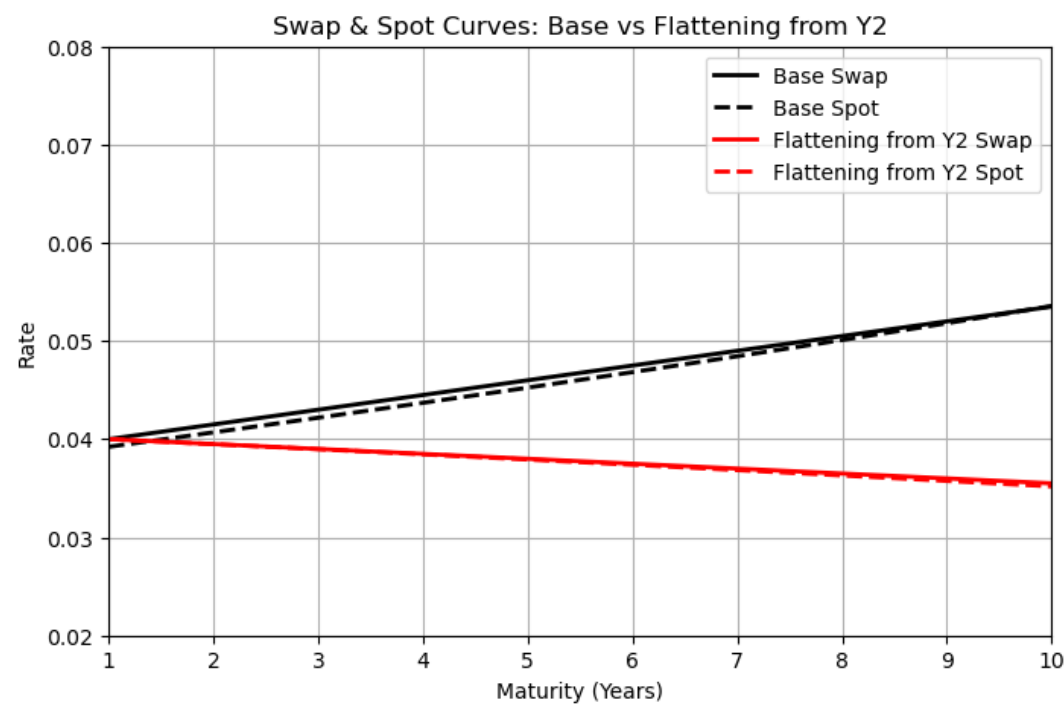
- Lower asset-side interest income from Short-Term investment
- Lower interest income for investment rollover
- Decreased value of Long-Term investment, no cash flow impact

Negative net cashflow effect given Short-Term Asset and Long-Term Liabilities

Cash Flow Impact on the Liability Side

- Lower interest expense for Short-Term liabilities
- Higher interest expense for Long-Term refinancing
- No cashflow impact from value depreciation of Long-Term liabilities

Interest Rate Scenario 5: Flattening the Curve Around 2Y



Cash Flow Impact on the Asset Side

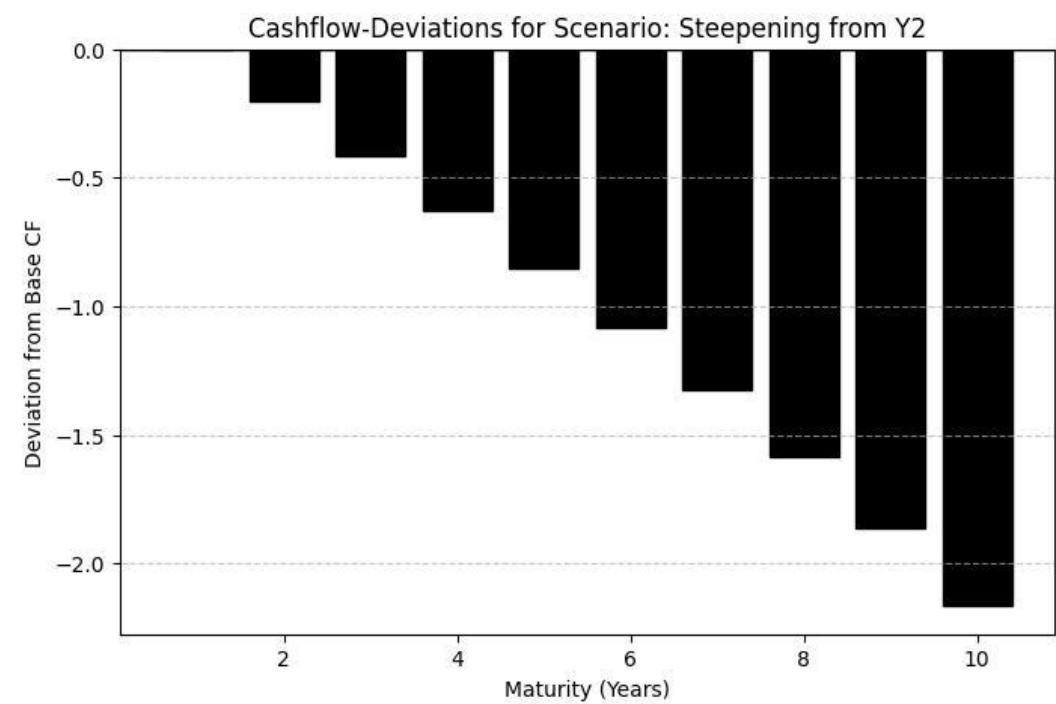
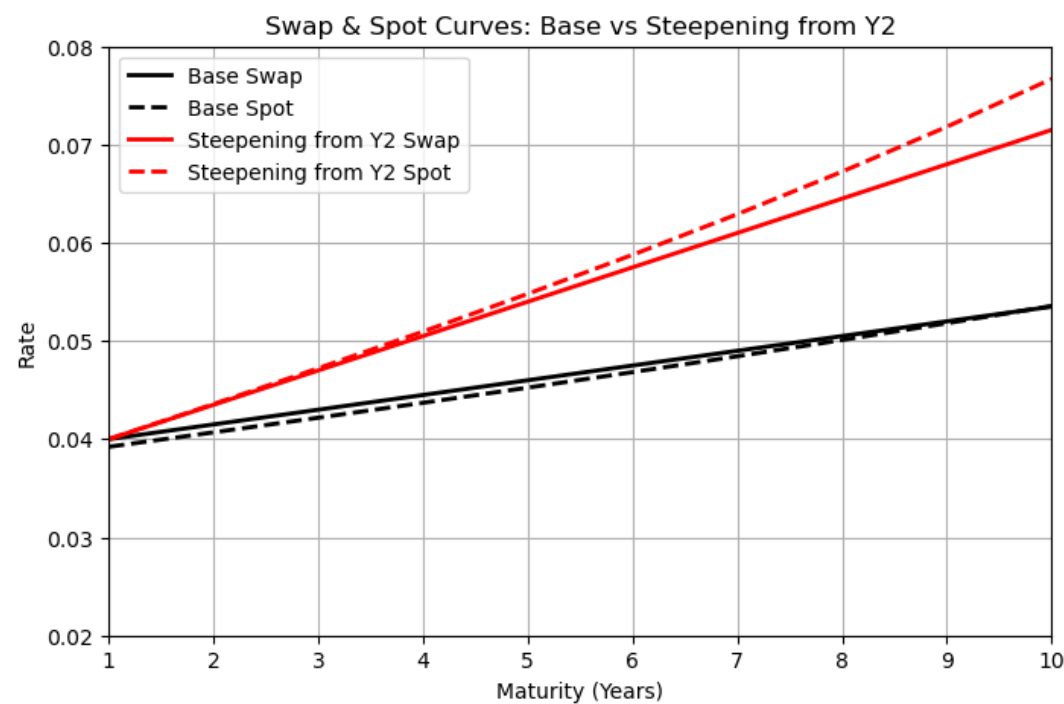
- Unchanged interest income from Short-Term investment
- Decreased interest income for investment rollover
- Increased value of Long-Term investment, no cash flow impact

Cash Flow Impact on the Liability Side

- Unchanged interest expense for Short-Term liabilities
- Lower interest expense for Long-Term refinancing
- No cashflow impact from value depreciation of Long-Term liabilities

Positive net cashflow effect given more pronounced impact on the long-term rate

Interest Rate Scenario 6: Steepening the Curve Around 2Y



Cash Flow Impact on the Asset Side

- Unchanged interest income from Short-Term investment
- Increased interest income for investment rollover
- Decreased value of Long-Term investment, no cash flow impact

Cash Flow Impact on the Liability Side

- Unchanged interest expense for Short-Term liabilities
- Higher interest expense for Long-Term refinancing
- No cashflow impact from value depreciation of Long-Term liabilities

Negative net cashflow effect given more pronounced impact on the long-term rate

Cash Flow at Risk Across Maturities

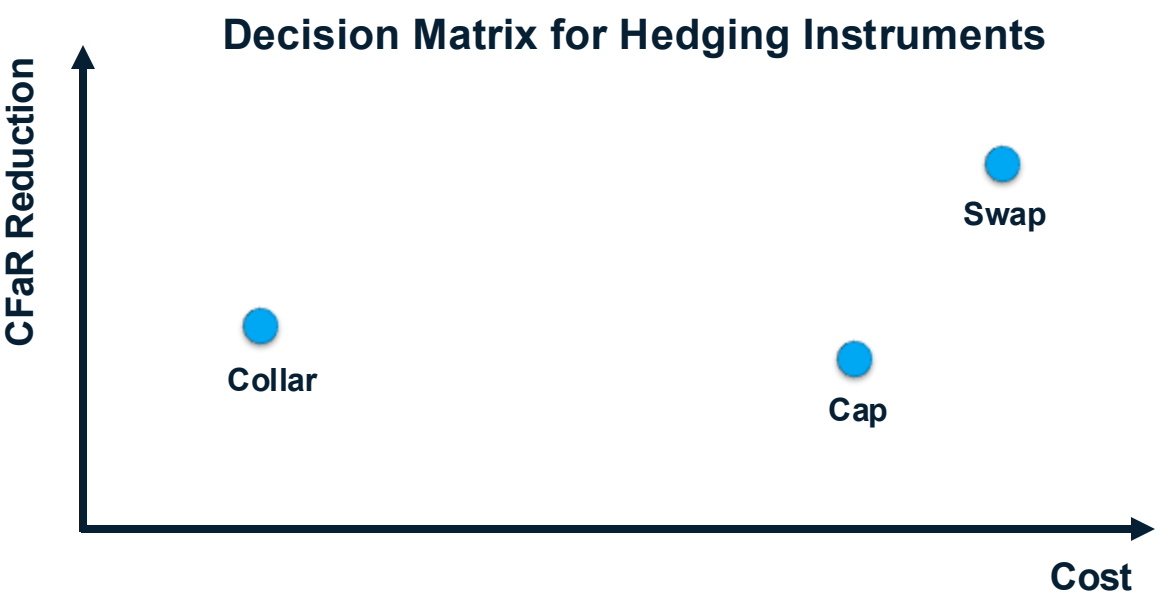
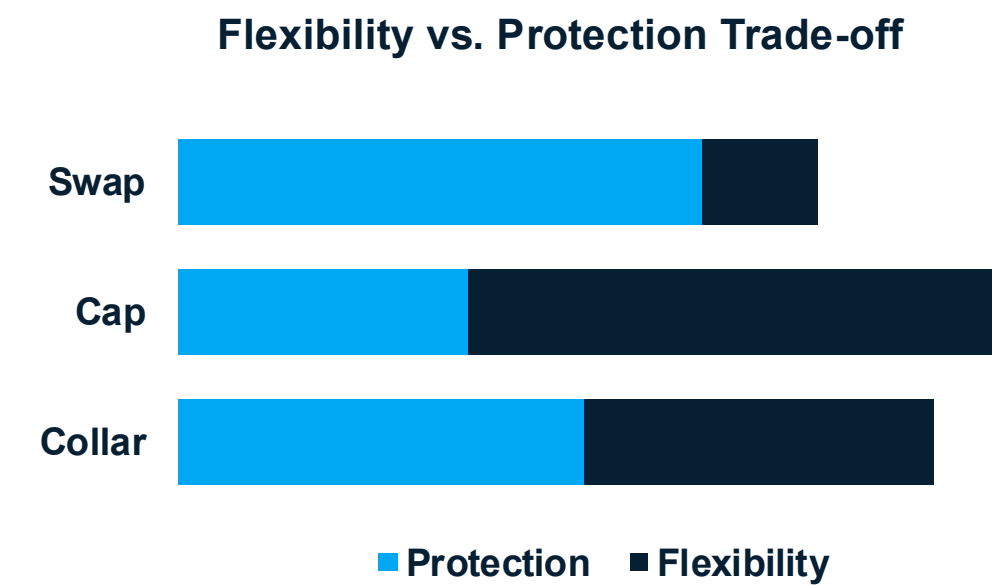
We assume one line item from the balance sheet, 100 Mio outstanding floating rate debt

Maturity (Years)	Expected CF (Base)	Parallel +100bp	Parallel -100bp	Flattening around 5Y	Steepening around 5Y	Flattening from Y2	Steepening from Y2	CFaR_95
1	-3.922071315	-5	-3	-4.8	-3.2	-4	-4	-1.027928685
2	-4.06919105	-5.153868231	-3.152366022	-4.748813087	-3.556234132	-3.949012989	-4.357638958	-0.983413395
3	-4.218476647	-5.31077314	-3.306674615	-4.696833049	-3.918508091	-3.897377295	-4.72234908	-0.945190478
4	-4.370247768	-5.471108868	-3.463229221	-4.644061212	-4.288441187	-3.845102439	-5.095862372	-1.007049476
5	-4.524859296	-5.635319649	-3.6223628	-4.590501233	-4.667888811	-3.792200038	-5.480189759	-1.07167788
6	-4.682708351	-5.803909806	-3.784443815	-4.648165745	-4.94320416	-3.738683855	-5.877712674	-1.176553606
7	-4.844242854	-5.977456136	-3.949883432	-4.706045337	-5.227705117	-3.684569829	-6.291307077	-1.368601488
8	-5.009972074	-6.156623411	-4.11914427	-4.764537945	-5.522691947	-3.629876096	-6.724515962	-1.57257075
9	-5.18047979	-6.342184007	-4.292751167	-4.82390701	-5.829878615	-3.574622991	-7.181796156	-1.791413329
10	-5.356440879	-6.53504306	-4.471304583	-4.884353307	-6.151431422	-3.518833035	-7.6688822	-2.028981535

"At 1-year maturity, 95% CFaR = €1m. Meaning our annual interest cost could be €5m instead of €4m. Treasury must plan an extra €1m liquidity buffer."

Hedging Strategy Analysis

Strategy Type	Basic Mechanism	CFaR Impact	Cost Strcture
Interest Rate Swap	Exchange floating-rate payments for fixed-rate payments	Reduces CFaR volatility	Fair value at inception
Interest Rate Cap	Provides protection when rates exceed the strike	Asymmetric CFaR protection	Premium payment required
Interest Rate Collar	Combination of cap and floor	Balanced CFaR reduction	Reduced or zero net premium



Calibration Choices: Tailoring CFaR to Treasury Governance

Time Horizon

- 1 Year: aligns with annual budget cycle, liquidity buffer sizing.
- 3–5 Years: captures refinancing and strategic debt mix.

Confidence Level

- 95% vs 99% Level: planning vs conservative buffer

Methodology

- Deterministic scenarios
- Historical simulation
- Monte Carlo

Market Inputs

- Swap curve: neutral market baseline
- Forward rates: market expectations
- Internal forecasts: Budget inputs

"Calibration makes CFaR flexible. By adjusting horizon, confidence, methodology, and inputs, the metric can be tailored to treasury governance and board risk appetite"

CFaR – A tool and a framework

CFaR is both a tool and a framework — it gives treasury an actionable liquidity risk number today, and a governance process to manage interest rate risk going forward

- **Liquidity focus:** quantifies worst-case cash shortfall, not mark-to-market swings
- **Board-ready metric:** simple output in € terms (95% confidence, 12m horizon)
- **Decision support:** evaluates hedging trade-offs (hedged vs unhedged)
- **Governance alignment:** ties risk reporting to board appetite and rating agency expectations

Recommendation

Model Configuration:

- Time Horizon: 12-month primary horizon with quarterly updates
- Confidence Level: 95% for management reporting, 99% for a more conservative view
- Scenario Framework: Equal weighting of all 6 scenarios as baseline approach
- Update Frequency: Monthly CFaR calculation with quarterly model recalibration

Calibration Recommendations:

- Start Simple: Begin with basic scenario analysis before advanced modeling
- Focus on Material Exposures: Prioritize largest interest rate exposures
- Validate Regularly: Quarterly backtesting against actual outcomes
- Document Assumptions: Clear audit trail for all model parameters

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Thank you !